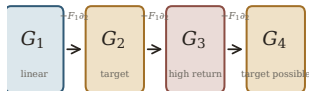


# Support thresholds in the invariant shear Picard chain



(a) Picard forest collapses to one chain



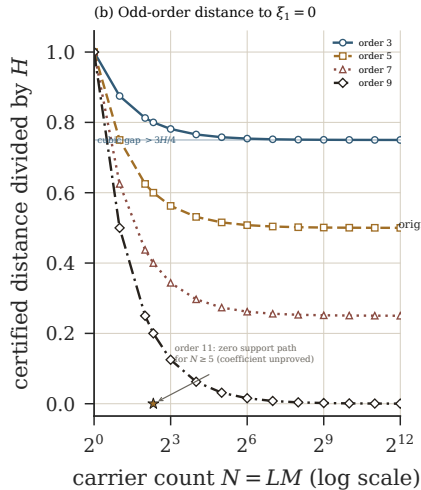
$$(\partial_t - \Delta_{12})G_n = -F_1 \partial_2 G_{n-1}$$

$u = (0, F, G) \cdot u^{[n]} = G_n e_3$  for  $n \geq 2$   
the second frequency  $m$  is conserved

Exact order-four identity:  $\frac{d}{dt}(2(G_1, G_3) + \|G_2\|_2^2) + 2(2(\nabla G_1, \nabla G_3) + \|\nabla G_2\|_2^2) = 0$ .

Formal scope: 24/24 checks; 32,771,750 exact support transitions through order 11; 323,216 Gaussian-integer convolution pairs for the energy regression.

Conclusion: cubic target return is impossible, but cubic high-frequency backtracking is essential. The packet is globally smooth in this shear class; this is not an arbitrary-data regularity proof.



(c) Target return and  $A^4$  energy

